

IoT in Industry: Architecture, Trends and Applications

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Lecture objectives

- Understand the layered architecture of Industrial IoT (IIoT), from device to cloud
- Position IIoT within Industry 4.0 and discuss its evolution through 2026
- Analyze connectivity, network determinism, edge computing and cybersecurity
- Discuss predictive/prescriptive maintenance and digital twins
- Review the global and Brazilian market landscape
- Analyze international and national application cases, with critical discussion

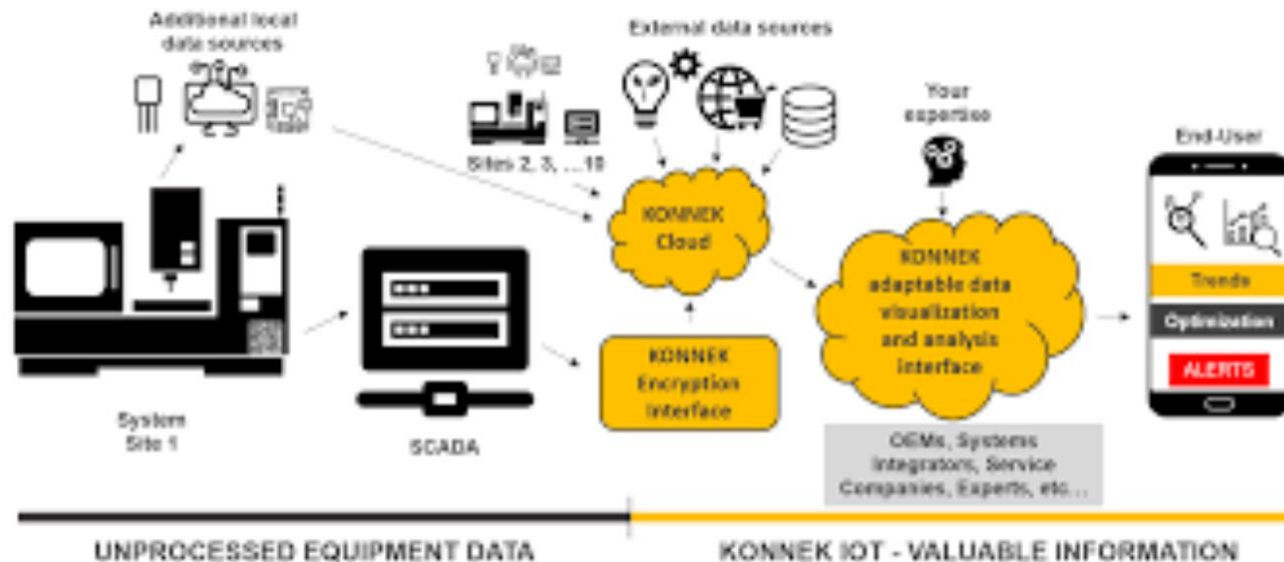
Smart factories

- Connected, automated, optimized and agile companies
- Use technology for better results, more efficient production and enhanced management
- Main benefits: reduced maintenance cost, energy savings, reduced waste
- Workforce productivity gains and improved service
- The foundation for revolutionary connectivity-enabled products and services



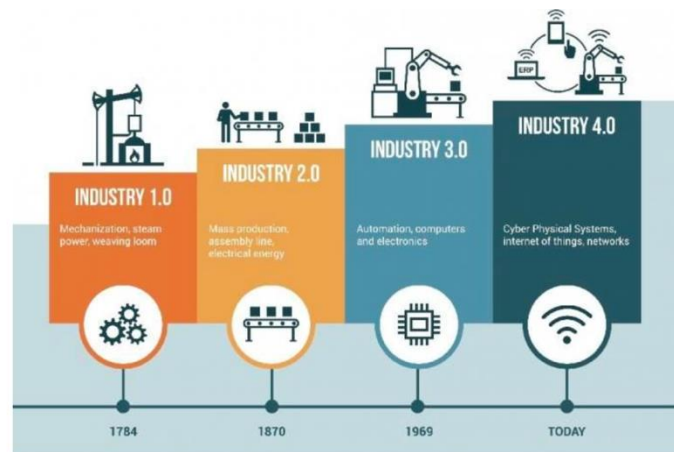
What is Industrial IoT (IIoT)?

- Applies IoT concepts to industrial environments: sensors, machines and connected systems
- Involves sensor data, machine-to-machine (M2M) communication and automation systems
- Core goal: turn raw operational data into actionable information for decision-making
- Sits at the intersection of Information Technology (IT) and Operational Technology (OT)



Industrial revolutions: 1.0 → 4.0

- Industry 1.0 (1784): mechanization, steam power, the weaving loom
- Industry 2.0 (1870): mass production, assembly line, electrical energy
- Industry 3.0 (1969): automation, computers and electronics
- Industry 4.0 (today): cyber-physical systems, internet of things, interconnected networks
- Each revolution drastically shortened the time between invention and large-scale adoption



Industry 4.0: the next phase

- Driven by: surging data volumes, computing power and low-power connectivity
- Internet of Things and Artificial Intelligence as central pillars
- New forms of human-machine interaction (augmented reality, natural interfaces)
- Advanced robotics and 3D printing
- Emerging trend: convergence of generative AI and IIoT (AIIoT) — language models interpreting sensor data and suggesting actions

IIoT vs. Industry 4.0 — the difference

- IIoT refers to sensor data, machine communications and automation systems
- Industry 4.0 covers the entire digital transformation of the sector (processes, business models, culture)
- IIoT is a technological enabler; Industry 4.0 is the broader strategic movement

The layered architecture of IIoT

- Device layer: sensors, actuators and controllers on the factory floor
- Edge layer: local pre-processing, filtering and real-time response
- Connectivity layer: wired and wireless networks linking devices and the cloud
- Cloud/platform layer: storage, heavy processing and analytics
- Application layer: dashboards, alerts and integration with enterprise systems (ERP/MES)

Smart sensors and industrial actuators

- Sensors act as the "eyes and ears" of machines, feeding data for decision-making
- Miniaturization and low power consumption expand the possible instrumentation points
- Sensors capable of operating in extreme conditions (temperature, vibration, dust)
- Mesh networks as a robust alternative in environments with irregular coverage
- Industrial wearables are gaining ground for worker safety and productivity

IT/OT convergence

- OT (operational technology): the network of operational processes and industrial control systems
- Includes human-machine interfaces, SCADA, distributed control systems and PLCs
- IT/OT convergence integrates automation and optimization with supply-chain visibility
- Historically, IT and OT evolved separately — today integration is a competitiveness requirement
- Core challenge: very different cultures, protocols and equipment life cycles

Industrial connectivity: protocols

- Industrial Ethernet and IP addressing brought IIoT down to sensor level on the factory floor
- MQTT: lightweight publish/subscribe protocol, widely used for IoT telemetry
- OPC-UA: interoperability standard across heterogeneous industrial systems
- CoAP: protocol designed for power- and processing-constrained devices
- Protocol choice depends on required latency, criticality and device capability

Wireless connectivity for IIoT

- LoRaWAN and NB-IoT: long range, low power, ideal for sparse telemetry
- Wi-Fi 7 and 5G-Advanced: high capacity and low latency for critical applications
- Hybrid networks combine wired, wireless and private technologies according to criticality
- 6G research projects (such as Brasil 6G) investigate massive connectivity for the next generation of industrial IoT
- Central design trade-off: range vs. power consumption vs. latency vs. data capacity

Network determinism

- In 2026, the central connectivity question is no longer "what's the peak bandwidth?"
- Applications like robotics, automation and safety systems require predictable latency
- Determinism now matters more than peak speed for real-time control
- Technologies such as 5G-Advanced and Wi-Fi 7 address this requirement
- Hybrid strategies (wired + wireless + private network) balance cost and reliability

Edge computing and Edge AI in industry

- Edge processing reduces latency and dependence on constant cloud connectivity
- Enables real-time response for critical process control
- Edge AI: artificial intelligence models running directly on edge devices
- Reduces the volume of data sent to the cloud, optimizing bandwidth and cost
- Requires balancing device computing capacity against model complexity

Cybersecurity in IIoT environments

- IT/OT convergence expands the attack surface of previously isolated systems
- Risks: legacy devices without updates, industrial protocols lacking native encryption
- Attacks on OT can cause direct physical impact, not just data loss
- Best practices: network segmentation, strong authentication, continuous monitoring
- Security is no longer optional — it becomes a design requirement from day one

Predictive maintenance: fundamentals

-50%

unplanned downtime

-25%

maintenance cost

- Uses sensor data and algorithms to predict failures before they occur
- Reduces unplanned stops and corrective-maintenance costs
- Depends on the quality and continuity of historical and real-time data
- Foundation for more advanced (prescriptive) maintenance models

From predictive to prescriptive

- Central trend for the industrial sector in 2026: prescriptive maintenance
- Systems don't just predict failure — they recommend the action and the ideal timing
- They factor in labor availability, spare parts and the production schedule
- The question shifts from "will this fail?" to "what should we do, and when?"
- Brings maintenance decisions closer to business outcomes

Digital Twins: concept and architecture

- A virtual replica of a physical asset, process or system, updated with real-time data
- Allows scenarios to be simulated before being applied to the physical environment
- Typical architecture: sensors on the physical asset → data sync → virtual model → analysis/simulation
- Used to test process changes, train operators and predict behavior
- A central component of the "smart factory"

Digital Twins: impact on maintenance cost

-55%

maintenance cost (digital twins)

- Market studies point to maintenance-cost reductions of up to 55% with digital twins
- They also reduce project time and rework on production lines
- They allow failure scenarios to be tested without risk to the real physical asset
- Upfront investment in instrumentation and modeling is the main limiting factor
- Adoption is accelerating in high-asset-value sectors (energy, heavy manufacturing)

Critical discussion: limits of digital twins

- The fidelity of a digital twin depends directly on the quality of the input data
- Keeping the model updated over the asset's life cycle is an underestimated challenge
- Integration with legacy systems is not always trivial
- Return on investment (ROI) is clearer for critical, high-value assets
- Open research question: how do we validate the reliability of a digital twin?

Economic impact of the Industrial Internet

US\$ 33 Tn

Estimated opportunity (McKinsey Global Institute)

- Historical analyses estimated the impact of the Industrial Internet in trillions of dollars
- Most impacted sectors: oil and gas, transportation, manufacturing, healthcare, energy
- Core benefits: reduced maintenance cost, energy savings, reduced waste

Global market landscape 2026

US\$ 188.5 Bn

Industry 4.0 market

39 Bn

Connected devices by 2030

US\$ 1 Tn

Annual IoT spending

- 41% of manufacturers prioritize investment in automation
- Industrial IoT is associated with productivity gains of up to 25%
- Advanced robotics can save up to US\$ 60K/year per worker replaced in repetitive tasks

Brazil landscape 2026

89%

Brazilian industries using advanced digital technology (PINTEC, 2024)

- Current challenge: many companies underuse the potential of tools they already own
- 2026 trends: prescriptive maintenance, mass adoption of IoT sensors, "all-in-one" platforms
- Factors restricting investment: taxation, operating cost, lack of system standardization
- Digitalization is no longer a competitive edge — it is now an operational necessity

Robots and automation in manufacturing

US\$ 60K/year

Estimated savings per worker replaced

- Robots and automated machines operate at minimum cost and maximum agility
- Used at multiple levels of the production chain, not only on the assembly line
- Smart data collection feeds inventory forecasting and predictive restocking analysis
- IoT sensors monitor environmental conditions and contamination risks

Case: DHL Smart Warehouse

- Warehouse digital-twin initiative for Tetra Pak's operations in Singapore
- Minimum 16% efficiency gain, with reduced congestion and better resource planning
- Live site access monitoring, with management alerts for controlled areas
- Real-time monitoring of environmental conditions (temperature)
- Full traffic-movement visibility for slotting optimization and material-handling equipment safety



Case: DHL Vision Picking (Augmented Reality)

- Use of augmented-reality glasses for guided order picking in the warehouse
- Login and navigation assisted by a scanner integrated into the wearable device
- Reduces picking errors and increases operator productivity



Case: Amazon Warehouse Order Picking Robots

- Autonomous mobile robots bring shelving units to human operators
- Reduces worker travel time and order-picking time
- Fleet coordination via a centralized warehouse-management system
- Scalability: the model is replicated across hundreds of distribution centers
- Example of large-scale collaborative automation (human + robot)

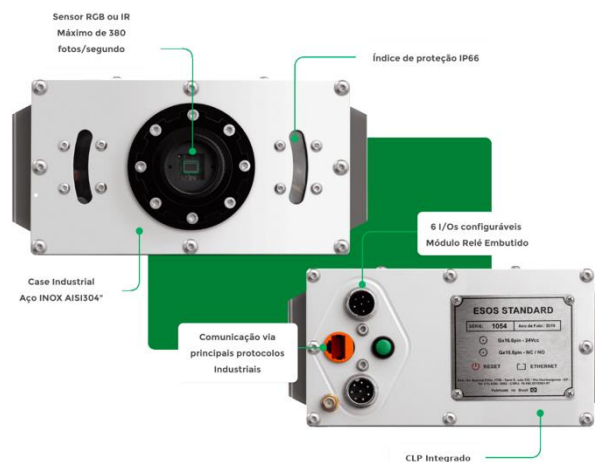


What these cases have in common

- Layered architecture: sensing/device → connectivity → processing → decision
- Real-time data connected to immediate operational decisions
- Convergence of physical automation (robotics) and digital systems (dashboards, AI)
- Focus on business metrics: efficiency, error reduction, safety, productivity
- Scalability as a design requirement from the outset

National case: Mvisia — Computer vision

- Brazilian company founded in 2012, specialized in computer vision and artificial intelligence
- Developed an entirely domestic industrial vision system
- RGB or infrared sensor with up to 380 frames per second
- Industrial-grade stainless-steel enclosure, IP66 protection rating
- Communication via the main industrial protocols, with an integrated PLC



National case: Birmind — AI-driven industrial optimization

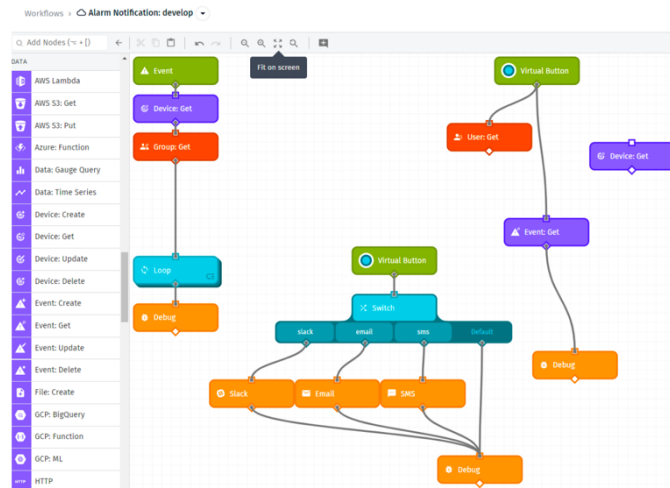
- Company based in Sorocaba, São Paulo state, Brazil
- Software that processes industrial data and correlates operational information
- Identifies potential waste-reduction opportunities in financial terms
- Ranks the best opportunities to optimize the factory
- Simulates scenarios through neural networks to support decision-making

National case: WEG



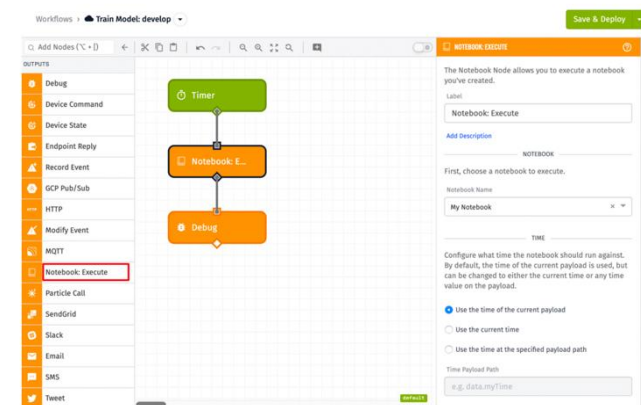
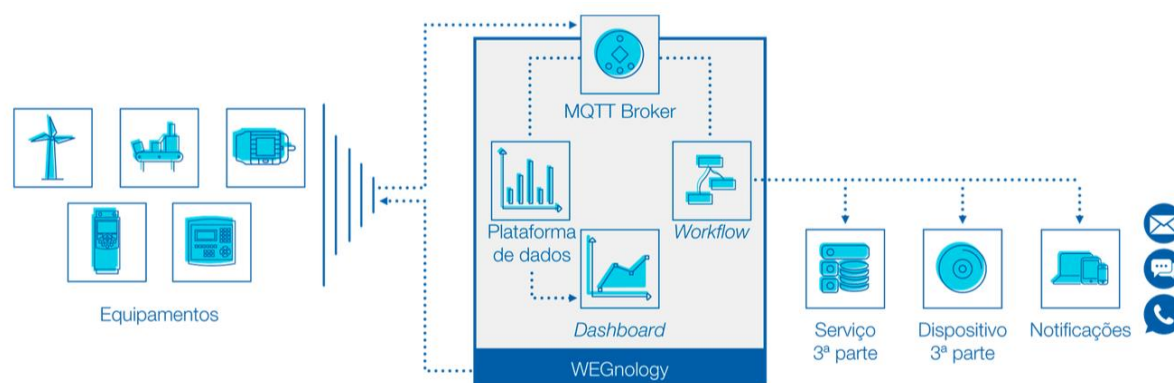
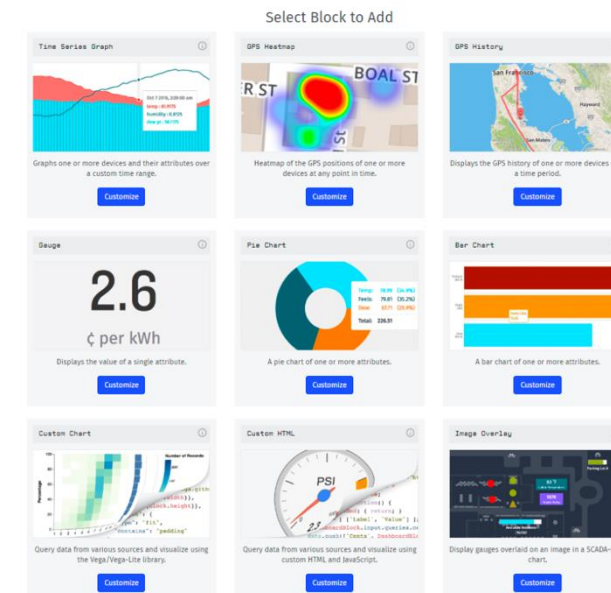
Low-code/no-code platforms for IIoT

- Drag-and-drop tools accelerate the development of IIoT solutions
- Allow fast adjustments as business rules evolve
- Lower the technical barrier for operations and engineering teams
- Example: visual workflow platforms integrating devices, cloud and notifications
- Market trend: essential for faster adoption in companies without large IT teams



National case: WEG / WEGnology

- WEGnology: cloud-based IoT platform for building connected solutions
- Collects, aggregates and visualizes data from sensors, motors, actuators and drives
- Drag-and-drop visual workflow accelerates development and business-logic adjustments
- Customizable dashboards, geolocation and correlation between variables
- Jupyter Notebook support for simulation, statistical modeling and machine learning
- WEG Motor Scan: diagnostic and monitoring solution for electric motors



Trends and open research questions

- Convergence of generative AI and IIoT (natural-language interpretation of operational data)
- Standardization of cybersecurity specifically for OT environments
- Validation and reliability of digital twins at industrial scale
- Next-generation deterministic connectivity (5G-Advanced, 6G) for critical applications
- Energy sustainability as a central design metric for IIoT systems

References, links and videos

- The next manufacturing revolution is here — Olivier Scalabre (TED) — <https://youtu.be/AyWtlwwEgS0>
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- The 7 Principles of the Industrial IoT (HPE) — <https://youtu.be/u3laXvjDiOE>
- DHL Smart Warehouse — <https://youtu.be/5zuysSUMGTc>
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- IoT For All — Six IoT Trends That Will Define 2026 — <https://newsletter.iotforall.com/p/six-iot-trends-that-will-define-2026>
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